Beyond History: Prediction of Olympic Medal Trends

Summary

Every four years, people from all over the world can feel the unique charm of the Olympics and cheer for their countries to win more medals. In this article, we established a series of predictive models to predict the number of medals each country will receive in the upcoming Olympic Games and their relationship with different events, while revealing the contribution of great coaches to a country's medal count and some of our original insights.

Firstly, we obtained a prediction model for the number of medals in the next Olympic Games by stacking random forests and gradient boosting, and compared it with the stacked prediction results of random forests and linear regression. We found that nonlinear factors have a greater impact on the prediction of Olympic medals than linear factors.

Next, we evaluated the probability of countries or organizations that have not won medals in history to obtain their first medal in the next Olympic Games.

Then, we explored the correlation between competition event settings and medal count through Lasso regression and VIF analysis, and found that the host event settings exhibited a concentrated distribution in competition events.

Finally,we take Lang Ping as an example and explore the impact of Lang Ping's coaching on the Chinese and American teams by establishing a double difference model. Research has shown that coaching Lang Ping has a significant positive impact on both the Chinese and American teams, providing strong evidence for the good coaching effect. At the same time, we searched for projects worth investing in good coaches in Cuba, Japan, and Italy, and used a difference in differences model to evaluate the positive impact of good coaches joining these projects.

Keywords: Olympics, Medals, Forecasting, Coach，Random forest,Cross validation

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1 Introduction

1.1 Background

Olympic Games are one of the main international event and also a matter of prestige for countries and therefore each country tries to give their best performance during the event. Despite a lot of hard work, many countries/player are unable to perform well during the events and grab medals whereas there are many countries which performs very well in the event and secures many medals[1].

The standings at the top of the table are often watched closely,but we should also pay attention to countries with relatively weaker Olympic capabilities at the current stage. It is gratifying that more and more countries have won their first Olympic medals, some even gold medals, in the Olympics. However, it cannot be ignored that there are still many countries that have not won any Olympic medals so far.

Figure 1：The LA28 Games

There are still three years left until the next Olympic Games - the Los Angeles Summer Olympics.The LA28 Games will be Los Angeles' third time hosting the Olympic Games (previously 1984, 1932) and the first time hosting the Paralympic Games[2]. The LA28 Games will represent the creativity, diversity and youthful energy of the Los Angeles region. Many countries are eager to try and achieve good results at the Olympics. Major institutions such as Nielsen have also begun to predict the results of the Olympics. At the same time, in order to achieve better results, some countries hire a large number of excellent coaches from various countries and promote the transfer of excellent coaches to their own countries in order to improve the level of their athletes.

We now have four available data file tables that record athlete information for all Summer Olympics from 1896 to 2024, complete national medal tables, host country lists, and statistics on the number of events for each sport. These pieces of information are crucial for us to predict the performance of each country in the next Summer Olympics, and are also an important tool for studying the contribution of a good coach to the number of medals won by a country. By studying these issues, we can provide decision-making support for the Olympic Committees of various countries and help them better prepare for the next Olympic Games.

1.2 Problem Restatement

By analyzing the above background, we summarize the tasks that need to be addressed as follows：

- **Task 1:** Establish a medal count model to predict the medal table for the 2028 Los Angeles Summer Olympics and provide a prediction interval for the results. Provide examples to indicate which countries will perform better and which countries will regress compared to the previous Summer Olympics.
- **Task 2:** Based on the model of Task 1, predict which countries are more likely to win their first Olympic medal at the next Olympics for those who have never won an Olympic medal before, and solve for their probability of winning a medal.
- **Task 3:** Determine the relationship between the Olympic events and the number of medals won by each country, and indicate which sports are more important for each country. Meanwhile, assess the impact of changes made by the host country to the event settings on the medal count of each country.
- **Task 4:** Estimate the contribution and impact of the "great coach" effect on the medal count of each country, and select three countries as examples to indicate in which sports they should hire "great coaches" to improve the performance of their athletes, and estimate the impact they will have.
- **Task 5：** Provide some original insights on the model in other aspects.

1.3 Our Work

To avoid complex descriptions and intuitively reflect our workflow, the flow chart is shown in the figure below:

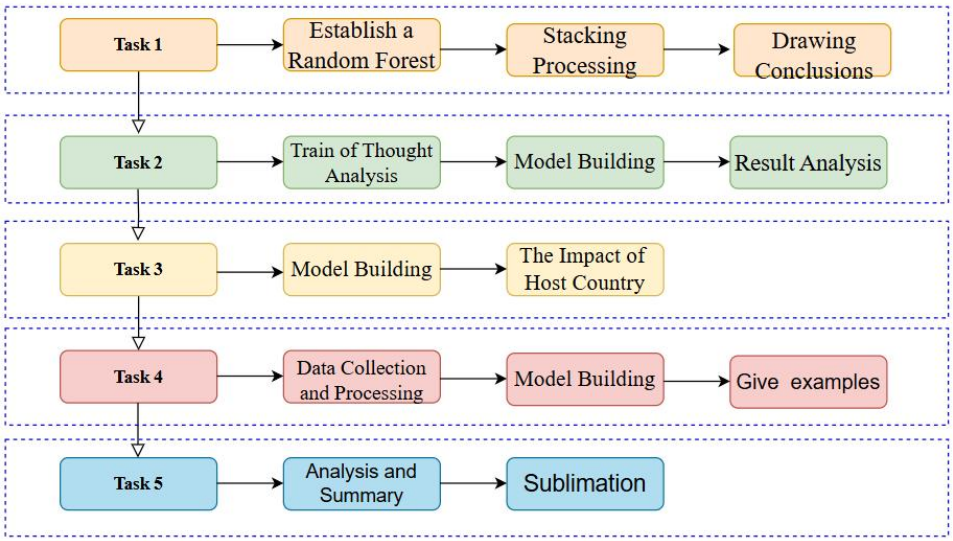

Figure 2：Our Work

2 Assumptions and Notations

Assumption 1: All data in this article are true and valid, without errors or false information.

Reason: The accuracy of data is the foundation of model correctness. We need to rely on accurate data to establish corresponding models, such as the Olympic medal prediction model.

Assumption 2: The Los Angeles Summer Olympics will not be significantly affected by political or natural factors, and Russia can participate normally.

Reason: Our model cannot predict the impact of some major emergencies (such as natural disasters) on the next Olympic Games. Considering Russia's outstanding performance in previous Olympic Games, we believe that whether Russia participates or not will have a significant impact on the number of medals won by each country. Therefore, it may be appropriate to assume that Russia participates normally, that is, it will not be unable to participate in the 2024 Paris Olympics due to special reasons.

Assumption 3: The countries or organizations participating in the Los Angeles Summer Olympics are composed of those that have participated in the 2020 or 2024 Olympics.

Reason: Considering that some countries do not participate in every Olympic Games, we believe that only countries that have participated in at least one of the previous two Summer Olympics will participate in the 2028 Los Angeles Summer Olympics.

Assumption 4: The Olympic level of each country will not experience a sudden rise or fall, meaning that participating countries will not make significant progress in traditional weaker events, nor will they experience significant setbacks in traditional advantageous events.

Reason: Only when the performance of each country shows relative stability, can the model prediction have stability and the prediction results are relatively reliable.

Assumption 5: The host country's reduction of certain events will not have a decisive impact on the number of medals and medal rankings of each country.

Reason: Taking the Los Angeles Summer Olympics as an example, boxing was reduced and five events including stick tennis and cricket were added. However, the changes in medals generated by the addition or subtraction of these events have a relatively small impact compared to the large number of medals generated by many other events. Therefore, it is not believed that these variables will have a significant impact on the medal count of certain countries.

In this work, we use the symbols in Table 1 in the model construction. Other none frequent-used symbols will be introduced once they are used.

Symbols	Definition
S_i	The total number of medals won by different countries in a certain event
γ_i	The degree of medal distribution in the most advantageous events of different countries
S_{total}	The total number of medals in a certain event
δ_i	The level of competition among the most advantageous projects in different countries
p_i	The probability of different countries winning medals
$Medals_{it}$	The number of medals won by women's volleyball teams from different countries in a certain Olympic Games

Table 1: Notations

3 Constructing the Medal Count Model Based on Random Forest and Gradient Boosting Stacking

3.1 Establish a Random Forest

3.1.1 Data Selection

The material provides relevant competition data for all Olympic Games from 1896 to the present. In the modeling process of this question, we only selected years when the Olympic Games have developed maturely and stably without major international emergencies for research, specifically the Olympic Games from 1992 to 2020. Because the prediction of Olympic medal numbers often does not need to be based on historical medal numbers over hundreds of years, but rather on the performance of various countries in recent years.

3.1.2 Data Input And Processing

We input six data sources into the random forest model[3], including the number of medals won in previous Olympic Games, the establishment of events in previous Olympic Games, the number of medals won in major events in previous Olympic Games, whether they are the host, the competition events in the next Olympic Games, and the host situation in the next Olympic Games, to predict the number of medals won by each country in the next Olympic Games.

The construction process of a random forest is also known as the construction process of a decision tree. For a new input data x, each decision tree Ti (x) will provide a predicted category yi. The final prediction of the random forest is:

$$\mathbf{y_f} \tag{1}$$

It is the majority vote of all decision tree predictions:

$$y_f = \frac{1}{n}\sum_{i=1}^{n} y_i \tag{2}$$

3.2 Stacking Processing And Drawing Conclusions

Generally speaking, the prediction results of a single random forest are not ideal. Therefore, we stack the prediction results obtained from the random forest with gradient boosting, input the same data (still the original six), and use cross validation to divide the training and validation sets. We round the predicted values to integers to obtain new predicted values and evaluate them. Below is a detailed process:

Firstly, train the random forest regression model to obtain the corresponding predicted values:

$$y_{RF} \tag{3}$$

Then, train the gradient boosting regression model to obtain the corresponding predicted values:

$$y_{GB} \tag{4}$$

Train a new regression model using the prediction results （y_{RF}、y_{GB}）of the first layer model as input, to fuse these two prediction results and obtain the final prediction result, which is:

$$y_{stacked} = \beta_0 + \beta_1 * y_{RF} + \beta_2 * y_{GB} \tag{5}$$

Among them, β_0、β_1 and β_2 are parameters in the second layer regression model, obtained through training data.

Afterwards, K-fold cross validation was used to divide the training data into K subsets, with K-1 subsets used for training and the remaining 1 subset used for validation, repeated K times.

Calculate the mean square error for all K validation predictions, which includes：

$$\mathbf{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_{true,i} - y_{pred,i})^2 \tag{6}$$

Where n is the number of data points, $y_{true,i}$ represents the true value of the data, and $y_{pred,i}$ represents the predicted value of the model.

Round the above data to obtain the predicted Olympic medal count for each country.

If you want to obtain a reasonable prediction interval based on the predicted values, you need to use the method of self sampling and compare it with the number of Olympic medals won by each country in 2024, in order to expand the predicted values into the prediction interval. The core idea of self sampling is to randomly select samples from the training data for repeated sampling, generating multiple "resampled" training sets. Train the model on these new training sets and obtain multiple prediction results.

These prediction results can be used to construct a prediction interval, which is obtained by sampling and estimating the predicted values multiple times to obtain the confidence interval of the predicted values. In this model, we constructed a 95% confidence interval, which includes:

Lower percentile:

$$y_{lower} = \mathbf{quantile}(0.025,\ y_1,\ y_2,\ \ldots\ldots,\ y_n) \tag{7}$$

Upper percentile:

$$y_{upper} = \mathbf{quantile}(0.975,\ y_1,\ y_2,\ \ldots\ldots,\ y_n) \tag{8}$$

Therefore, the final prediction interval is:

$$[\ y_{lower}\ ,\ y_{upper}\ ] \tag{9}$$

Based on the above results, the final medal table is as follows. Due to limited space, we only selected the top ten countries on the gold medal table, and attached the upper and lower bounds of the predicted medal count range for these countries

Country	Gold	Silver	Bronze	Total	Upper Bound	Lower Bound
China	33	46	29	108	83	111
United States	25	15	78	118	92	120
Russia	20	23	13	56	40	64
Australia	20	15	20	55	39	63
France	18	31	2	51	35	59
South Korea	18	7	14	39	25	49
Netherlands	11	10	27	58	42	66
Germany	11	7	30	48	33	57
Canada	11	6	11	28	15	39
New Zealand	10	5	1	16	11	23

Table 2: Los Angeles Olympics (2024) Final Medal Table – Gold Medal Top 10 Countries

From the predicted results in the table above, if ranked by the number of gold medals, although the number of gold medals in both China and the United States has decreased compared to the 2024 Paris Olympics, China's ranking will surpass the United States to become the first place. Other countries that have risen in rankings include South Korea, Germany, Canada, New Zealand, Denmark, etc. Correspondingly, countries such as Japan, Great Britain, and Italy will decline in rankings, while countries such as Australia and France will remain unchanged in rankings (note: as Russia did not participate in the 2024 Paris Olympics, its predicted rankings for 2028 will not be compared).

Of course, if ranking is based on the total number of medals, the rankings of each country will also change. For example, the United States will still rank first, while the Netherlands' ranking will rise to third place.

The following chart shows the comparison of medal counts between several world sports powerhouses in 2024 and 2028:

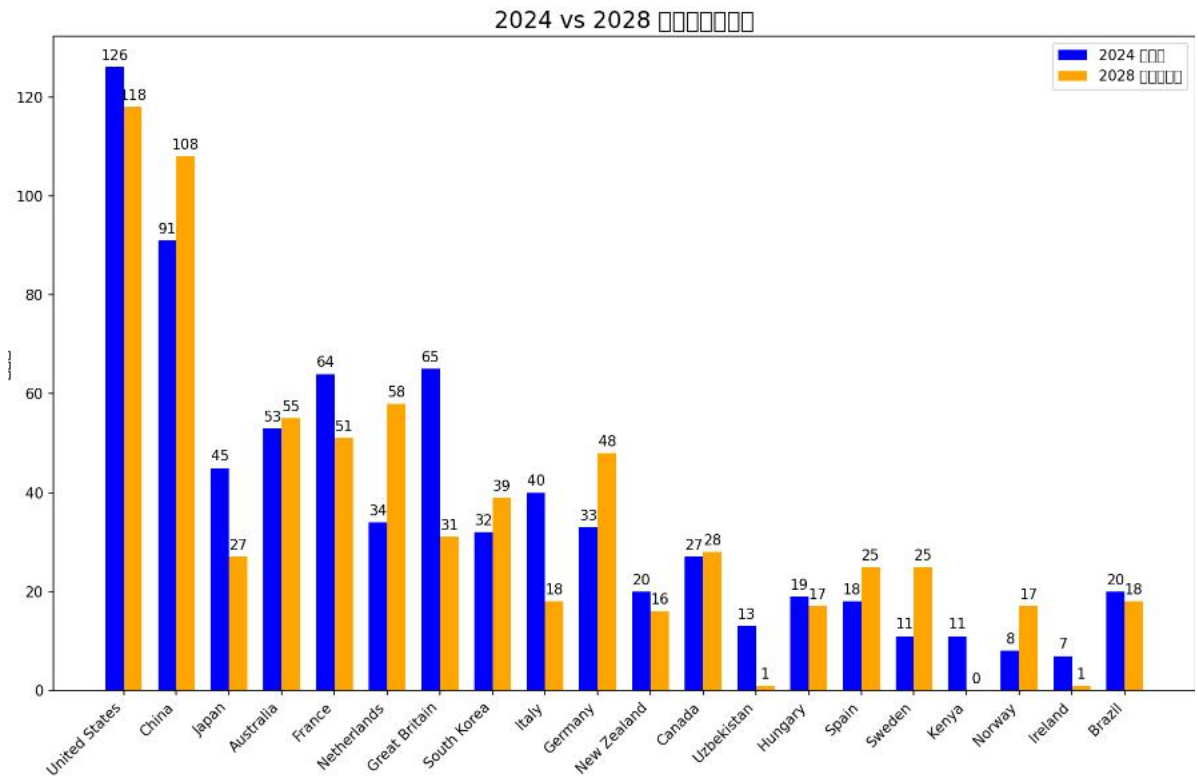

Figure 3 :Medal table

It should be noted that the prediction results of the model we used in this question are not absolutely accurate, and there is a certain deviation in its predictions for gold, silver, and bronze medals. The mean square error of the three is calculated as follows:

Gold	0. 04293935369632581
Silver	0. 04527345164795364
Bronze	0. 048452654030343274

Table 3: Mean squared error

4 Predicting Which Countries Will Win Medals for the First Time in the Next Olympic Games and Their Probability

4.1 Train of Thought Analysis

Currently, there are still over 60 countries that have not won Olympic medals. In order to predict the probability of countries that have never won an Olympic medal in history winning their first medal at the next Olympics, we believe that these countries can only win medals in their most advantageous events at the next Olympics. This is because considering that these countries have participated in many Olympic Games but have never won a medal, the probability of winning two or more medals simultaneously in the next Olympic Games can be ignored.

Strictly speaking, any country that has not yet won an Olympic medal has the possibility of winning their first medal in the next Olympics, and the difference lies only in the probability. So, we analyze the comprehensive Olympic strength of these countries and quantify them as this indicator. Considering countries that have only won one Olympic medal in history, we believe that their basic situation in the Olympics is similar to those countries that have never won an Olympic medal in history. Therefore, from the relevant data of countries that have only won one Olympic medal, a threshold score can be derived, which can be used to infer the probability of countries that have not yet won an Olympic medal winning their first medal in the next Olympics.

4.2 Model Building and Result Analysis

Based on the above analysis, we start with the scores of different countries to calculate the probability. We believe that Score should consider five factors, namely:

① Political stability in the target country:

$$\alpha_i \tag{10}$$

② Target country GDP:

$$\beta_i \tag{11}$$

③ The degree of medal allocation for the most advantageous events in the target country:

$$\gamma_i \tag{12}$$

④ The level of competition for the most advantageous projects in the target country:

$$\delta_i \tag{13}$$

⑤ Comprehensive strength of athletes from target countries:

$$\varepsilon_i \tag{14}$$

These five factors collectively affect the total score. We believe that the higher the score, the stronger the country's overall strength in the Olympics, and the more likely it is to win medals in the next Olympics.

Assign values to different aspects of countries that have not won medals by consulting materials[4][5][6], with the specific range of values being: α_i {0 , 1}， β_i {0 , 0.5 , 1}， ε_i {0 , 1}.

By consulting relevant materials[7] and the data provided in this question, determine the most advantageous projects corresponding to these countries and the degree of competitiveness of these projects. The definition of the most advantageous project for the target country refers to the event in which the country has participated the most in previous Olympic Games. If the medals in the most advantageous event are consistently won by only a few countries, the probability of the target country winning medals in that event is considered low. If the medals in the most advantageous event are evenly distributed, the probability of the target country winning medals in that event is considered high.We use the Herfindahl Hirschman Index to characterize the degree of medal allocation in a project.Its definition is as follows:

$$\mathbf{HHI} = \sum_{i=1}^{n} \left(\frac{S_i}{S_{total}}\right)^2 \tag{14}$$

Among them, n is the number of countries that have won medals. The degree of medal allocation we are seeking:

$$\gamma_i = 1 - \mathbf{HHI} \tag{15}$$

As for the level of competition in a certain project, if n is the number of countries participating in the project, if a country has more countries participating in its most advantageous project, then the competition in that project will be relatively fierce (which means that these countries will have more difficulty winning medals), otherwise it will not be intense. Competition intensity:

$$\delta_i = \frac{1}{n} \tag{16}$$

Through experiments, we have determined appropriate parameters and obtained the following calculation method for Score:

$$\mathbf{Score} = 0.2 \cdot \alpha_i + 0.2 \cdot \beta_i + 0.15 \cdot \gamma_i + 0.15 \cdot \delta_i + 0.3 \cdot \varepsilon_i \tag{17}$$

After scoring each country that has never won an Olympic medal in history separately, we will now search for those countries that have won a total of medals in the next Olympic Games, and define these countries as those with the ability to win medals. Calculate the Score of these countries, take the average to obtain the threshold score, and record it as T.

Finally, the probability of each target country winning a medal in the next Olympic Games is obtained as follows:

$$p_i = \frac{Score}{T} \tag{18}$$

Based on the above results, the probability distribution table of countries that have never won Olympic medals in history winning medals in the next Olympic Games is obtained. Due to limited space, we have only selected ten countries (listed alphabetically as the top ten) and presented their probabilities.

Country	Probability
Andorra	0.232258065
Angola	0.361290323
Aruba	0.309677419
Bangladesh	0.206451613
Benin	0.309677419
Bhutan	0.15483871
Belize	0.15483871
Bolivia	0.283870968
Brunei	0.361290323
Central African Republic	0.232258065

Table 4: The probability of different countries winning medals

5 Judging the Importance of Different Sports to a Country Based on Lasso Regression Model

5.1 Model Building

The coefficient size of Lasso regression can help us determine the importance of different projects to various countries. The larger the absolute value of the coefficient, the greater the contribution of the event to the total number of medals. So our goal is to solve the coefficients of each event.

For the dataset given in this question, the objective of the Lasso regression model is to minimize the following objective function:

$$\boldsymbol{\beta}_{\mathbf{final}} = \mathbf{argmin}\left(\sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2 + \mu \sum_{j=1}^{p} |\beta_j| \right) \tag{19}$$

of which：

y_i is the total number of medals won by the i-th country;

x_{ij} is the number of medals won by the i-th country in the j-th event;

β_j is the coefficient of each event, representing the degree to which the event affects the total number of medals;

μ is the regularization parameter of Lasso regression, used to control the sparsity of coefficients.

Summarize the above results and obtain the Lasso regression coefficient table for different sports events as follows. Due to space limitations, we only selected ten sports events (listed alphabetically as the top ten) and presented their corresponding Lasso regression coefficients.

Sports Events	Lasso Regression Coefficient
Archery	0.498487964
Artistic Gymnastics	0.5104201
Artistic Swimming	0.344008086
Athletics	2.627979771
Badminton	0.403088074
Baseball	1.280359248
Baseball	0.901522615
Basketball	1.583747366
Beach Volleyball	0.212873883
Boxing	0.576613359

Table 5：Lasso regression coefficient distribution table

In the above table, the larger the coefficient of a project, the greater its importance to each country. By arranging the coefficients in descending order, one can identify which movements are most important for each country. The top three sports are Swimming, Athletics, and Hockey. Compared to other sports, the importance of these sports is clearly higher.

5.2 Host Country's Event Setup

In order to investigate the impact of the host country's event settings (newly added and deleted sports) on the final results, we calculated the average Lasso regression coefficient of each newly added event in the host country for each Olympic Games, and compared it with the average Lasso regression coefficient of all events in the Lasso regression coefficient table obtained in 5.1 to determine whether the newly added event in the host country is an important event for the country.

Average regression coefficient for all projects	Average regression coefficient of additional projects in the host country
0.560367728	0.791873683

Table 6：Comparison of coefficients

It can be seen that the average Lasso regression coefficient for all competition events is 0.560367728, which is significantly lower than the average Lasso regression coefficient for the events added by the host country in each Olympic Games. This indicates that the competition events added by the host country are often important events for the country, and adding these events to a certain extent enables the host country to achieve better results and win more Olympic medals. Through this model, it can also reflect the significant impact of the host country effect on a country's medal count.

6 Taking Lang Ping as an Example, Exploring the Impact of Great Coach Effect Based on the Double Difference Model.

6.1 Data Collection and Processing

We take Lang Ping as an example to study the coaching effect. Lang Ping became the coach of the US team in 2008 and the Chinese team in 2016 in the history of the Olympic Games. Based on Lang Ping's coaching experience, we have selected the award data for women's volleyball events at various Olympic Games from 1992 to 2024 provided in the materials.

Due to the fact that there are only four types of awards in the women's volleyball event at each Olympic Games: no medal, bronze medal, silver medal, and gold medal. In order to facilitate the calculation of the model, we assigned values to the winning data of the women's volleyball event: {No medals won: 0, bronze medals won: 1, silver medals won: 2, gold medals won: 3}.

In order to establish the subsequent double difference model, we need to add four columns of data based on the winning data of women's volleyball events at the Olympic Games of various countries from 1992 to 2024, namely:

① `Treat_USA`, which indicates whether it is the US team, with a value of 1 and not 0.

② `Treat_China`, which indicates whether it is a Chinese team, with a value of 1 and not 0.

③ `Post_USA`, which indicate whether it is the year after Lang Ping became the Olympic coach of the US team, which is 1, not 0.

④ `Post_China`, which indicate whether it is the year after Lang Ping became the Olympic coach of the Chinese team, which is 1, not 0.

6.2 Model Building and Result Analysis

The core idea of the double difference model is to estimate the impact of intervention by comparing the differences in changes between the intervention group (i.e. the group affected by policies or events) and the control group (i.e. the group not affected) before and after intervention.

By inputting the processed award data of women's volleyball events from 1992 to 2024 Olympic Games in various countries, we can obtain the following formula:

$$\boldsymbol{Medals_{it} = \alpha + \lambda_t + \gamma_i + \beta_1 \cdot Post_{USA} + \beta_2 \cdot Treat_{USA} + \beta_3 \cdot (Treat_{USA} \times Post_{USA}) + \beta_4 \cdot Post_{China} + \beta_5 \cdot Treat_{China} + \beta_6 \cdot (Treat_{China} \times Post_{China}) + \epsilon_{it}} \quad (20)$$

Among them:

$\boldsymbol{Medals_{it}}$, which represents the number of medals won by the i-th country's women's volleyball team in the t-th year.

λ_t , which is the annual fixed effect.

$\boldsymbol{\gamma_i}$, which is the fixed effect of the country.

$\boldsymbol{\epsilon_{it}}$, which is the error term.

$\boldsymbol{Post_{USA}}$, which is a dummy variable representing before and after Lang Ping became the Olympic coach of the US team.

$\boldsymbol{Post_{China}}$, Which is a dummy variable representing before and after Lang Ping became the Olympic coach of the Chinese team.

$\boldsymbol{Treat_{USA}}$, which is a dummy variable, indicating whether it is the US team or not.

$\boldsymbol{Treat_{China}}$, which is a dummy variable indicating whether it is the Chinese team.

$\boldsymbol{\beta_i}$ (i=1, 2, 3, 4, 5, 6) , which is the parameter, solved through the model, where $\boldsymbol{\beta_3}$ and $\boldsymbol{\beta_6}$ are the interaction coefficients of the Chinese team and the American team, respectively. We reflect Lang Ping's good coaching effect through interaction coefficient β_3 and β_6. If the coefficient is positive, it indicates that Lang Ping has had a positive impact on both teams and there is a good coaching effect. Simultaneously calculating the p-values as well as β_3 and β_6, if the p-value is less than 0.05, it is considered that Lang Ping's good coaching effect is significant.

After running the model, we obtained the following table results:

$\boldsymbol{\beta_3}$	$\boldsymbol{\beta_6}$
0. 6585	0. 8949

Table 7: Coefficient of interaction term

$\boldsymbol{p_3}$	$\boldsymbol{p_6}$
0.0353	0.0279

Table 8: Comparison of p-values

$\boldsymbol{\beta_3}$ = 0. 6585, this means that after Lang Ping becomes the Olympic coach of the US women's volleyball team, the team can theoretically add 0.6585 bronze medals, with a p-value of 0.0353, which is less than 0.05. Therefore, it can be considered that Lang Ping's influence on the US women's volleyball team is significant.

$\beta_6 = 0.8949$, this means that after Lang Ping became the Olympic coach of the Chinese women's volleyball team, the team theoretically gained 0.8949 bronze medals, with a p-value of 0.0279, which is less than 0.05. Therefore, it can be considered that Lang Ping's influence on the Chinese women's volleyball team is significant.

Lang Ping contributed 1.4534 bronze medals to the total medal count of the Chinese and American teams.

Therefore, Lang Ping has a positive impact on the medal increase of both teams, with a total contribution of positive.

6.3 Analyzing Great Coach Investments in Cuba, Japan and Italy

6.3.1 Definition of Projects that Require Investing in Great coach

Every country participates in numerous events at the Olympics, so which part of the events is most worth investing in as a good coach to improve performance? We believe that if certain countries' projects have potential but have only recently shown weak performance, then these projects are worth investing in as good coaches for that country.

6.3.2 Reasons for Choosing Cuba, Japan and Italy

Although Cuba, Japan, and Italy perform well in some traditional Olympic events, many other events often fail to showcase their full potential.

Cuba excels in sports like boxing, volleyball, and athletics, but struggles in swimming and badminton due to a lack of high-level coaches and outdated training methods. This limits their ability to reach the full potential of their athletes. Japan has strengths in judo and gymnastics, but lacks top coaches in swimming and athletics, hindering progress compared to countries like Europe, the US, and Australia. Italy, known for cycling and football, also has potential in events like swimming and athletics but lacks professional coaching, which affects their performance in technical training and overall competitiveness.

So we believe that if good coaches are introduced for some of the events in these three countries, it will significantly increase the number of medals they win in these events.

6.3.3 Data Analysis and Project Selection

We have selected the award data for all events from the 1992 to 2024 Olympic Games in Cuba, Japan, and Italy. Due to the fact that there are only four types of individual event awards in each Olympic Games: no medal, bronze medal, silver medal, and gold medal. For the convenience of model calculation, we assigned values to the award data of various projects in Cuba, Japan, and Italy: {No medals won: 0, bronze medals won: 1, silver medals won: 2, gold medals won: 3}.

We believe that if certain events in Cuba, Japan, and Italy have won at least two medals before the 2016 Olympics but have not won a single medal in the 2016 and subsequent Olympics, then these events have potential but have shown weak recent performance and are worth investing in as good coaches for the country.

Based on this criterion, we have selected projects from Cuba, Japan, and Italy that are worth investing in as good coaches, and according to the model established in 6.2, we have calculated the theoretical number of medals that these projects can win in this project at the 2028 Olympics after investing in good coaches, as shown in the table below:

Team	Sport	Number of Medals
Cuba	Baseball	0. 737661138
Cuba	Canoeing	0. 142096372
Cuba	Cycling	0. 168487482
Cuba	Fencing	0. 078107942
Cuba	Swimming	0. 31402347
Cuba	Volleyball	0. 394906106
Cuba	Weightlifting	0. 472196675
Italy	Basketball	0. 682302398
Italy	Canoeing	0. 737095506
Italy	Cycling	0. 707470591
Italy	Diving	0. 163264858
Italy	Football	0. 309097389
Italy	Gymnastics	0. 411982112
Italy	Water Polo	0. 744285318
Japan	Baseball	0. 574355767
Japan	Canoeing	0. 346751973
Japan	Cycling	0. 44504656
Japan	Football	0. 85952488
Japan	Gymnastics	0. 657243308
Japan	Shooting	0. 534990657
Japan	Softball	0. 575461173
Japan	Synchronized Swimming	0. 041370344
Japan	Taekwondo	0. 246744215
Japan	Tennis	0. 039432718
Japan	Volleyball	0. 303203941

Table 9：Project Selection

7 Model Evaluation and Improvement

7.1 Regarding Task 1

In the medal prediction model, we used a stack of random forest and gradient boosting, but both models are mainly used to deal with nonlinear problems, so this stack may be slightly insufficient in dealing with linear parts. Therefore, we believe that stacking random forests with models that focus on handling linear relationships, such as stacking random forests with linear regression, stacking random forests with logistic regression, etc., can be added to compensate for the shortcomings of stacking random forests and gradient boosting. At the same time, we compare with stacking random forests and gradient boosting to explore the different effects of linear and nonlinear factors on medal prediction. Below are only partial predicted values and errors for the stack of random forest and gradient boosting:

Country	Gold	Silver	Bronze	Total
United States	48	16	29	93
Japan	9	21	19	49
France	14	29	3	46
China	31	12	2	45
Russia	15	17	12	44
Germany	14	5	22	41
Spain	16	13	10	39
Australia	10	14	13	37
Netherlands	11	14	9	34
Argentina	3	11	15	29
Italy	7	12	7	26
Brazil	11	11	3	25
Cuba	18	6	1	25
South Korea	3	7	15	25
Croatia	6	7	11	24
Hungary	7	8	9	24
Canada	6	6	7	19
Great Britain	5	5	9	19
New Zealand	12	0	6	18
Serbia	5	5	8	18
Nigeria	2	11	3	16
Norway	4	3	9	16
Denmark	9	1	5	15

Table10:Stacking Random Forests and Linear Regression

From an error perspective, the stacking of random forests and gradient boosting results in larger mean square errors for both gold and silver medals. At the same time, compared with previous years' data, some anomalies in data predictions can be clearly found, such as China only being able to win two bronze medals, which is significantly different from the winning situation in previous Olympic Games.

Therefore, in our context, the prediction performance of the stack of random forest and gradient boosting is better than that of the stack of random forest and gradient boosting. Based on this result, we believe that the influence of nonlinear factors is stronger than that of linear factors in the prediction of Olympic medals.

7.2 Regarding Task 2

In predicting the probability of countries that have not yet won medals winning their first medal, to simplify the problem, we assume that countries or organizations that have not won medals in the history of the next Olympics may only win medals in their most advantageous events. However, in reality, countries or organizations that have not won medals in the history of the Olympics may not only win medals in their most advantageous events, but also have the possibility of winning medals in some relatively weaker events, although the probability may be lower.

To take these projects into consideration, an analysis can be conducted based on the specific strengths of the target countries in participating in the projects.We have improved the parameters in the Score calculation process and introduced Monte Carlo method to simulate the probability of the target country winning its first medal in the next Olympic Games:

$$\mathbf{p(win)}=\frac{x}{y} \tag{21}$$

Where x is the number of simulations for winning medals, and y is the total number of simulations.We believe that the results obtained in this way will be more in line with the actual situation.

7.3 Regarding Task 3

In exploring the relationship between medal count and event design, we originally discussed whether the host country's events were important events for the country. On this basis, we avoid analyzing the changes in the events caused by the host country and further explore whether the establishment of the host country itself (using 1 to indicate that the country is the host country and 0 to indicate that the country is not the host country) is related to the number of medals, and analyze which events the changes in the number of medals caused by the host country information mainly focus on. The specific steps are as follows:

1. Calculate the Pearson correlation coefficient between the host country's information and the total number of national medals.

Pearson correlation coefficient is a statistical measure of the strength and direction of the linear relationship between two sets of data. It is used to evaluate the linear correlation between

two variables, that is, whether the variables show the same trend when changing, or the trend of one variable changing when the other variable changes. The Pearson correlation coefficient ranges between -1 and+1.

+1 represents complete positive correlation, meaning that there is a completely linear relationship between two variables and they always change in the same direction.

-1 represents complete negative correlation, meaning there is a completely linear relationship between two variables and they always change in reverse.

0 indicates that there is no linear relationship or obvious linear dependence between two variables.

The formula is as follows:

$$\mathbf{r} = \frac{\sum(\mathbf{X_i}-\overline{\mathbf{X}})(\mathbf{Y_i}-\overline{\mathbf{Y}})}{\sqrt{\sum(\mathbf{X_i}-\overline{\mathbf{X}})^2(\mathbf{Y_i}-\overline{\mathbf{Y}})^2}} \tag{22}$$

The X variable represents the host information, the Y variable represents the number of national medals, and $\overline{X}$、$\overline{Y}$ are the mean values of X and Y, respectively.

By calculation, we obtained a Pearson correlation coefficient of 0.271196178.

2.Conduct a t-test to determine if the correlation between host information and the total number of national medals is significant.

The p-value we obtained is 4.39E-278, which is less than 0.05, indicating a significant correlation between the host country's information and the total number of national medals.

3. Determine which events the changes in the number of medals caused by the host country's information are mainly concentrated in by calculating the density of medal changes.

Partial results are as follows:

Country	Competition item	Year	Medals	Host country	Total	Change	Medal variation density
Afghanistan	Taekwondo	2008	1	0	1	1	1
Albania	Wrestling	2024	2	0	2	2	1
Algeria	Artistic Gymnastics	2024	1	0	3	1	0.333333333
Algeria	Athletics	2000	4	0	5	3	0.6
Algeria	Athletics	2024	1	0	3	1	0.333333333
Algeria	Boxing	1996	2	0	3	1	0.333333333
Algeria	Boxing	2024	1	0	3	1	0.333333333
Algeria	Judo	2008	2	0	2	2	1
Argentina	Basketball	2004	12	0	49	12	0.244897959
Argentina	Boxing	1996	1	0	20	1	0.05
Argentina	Cycling	2008	2	0	51	2	0.039215686
Argentina	Cycling BMX Freestyle	2024	1	0	19	1	0.052631579
Argentina	Football	1996	18	0	20	18	0.9

Visualize the above image:

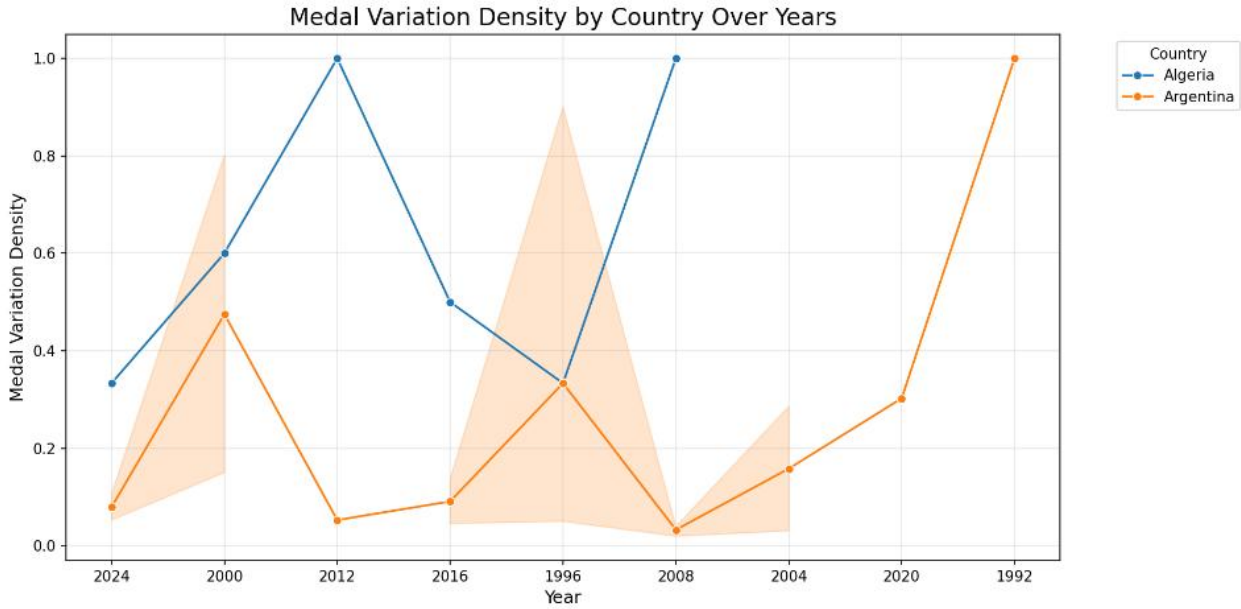

Figure 4 :Medal density

From the table, it can be seen that the changes in the number of medals caused by host information are mainly concentrated in a small number of events, such as Judo, Wrestling, Taekwondo, etc., with clear directionality.

7.4 Regarding Task 4

In exploring the issue of good coaching effects, we have selected specific countries to investigate the projects and impacts they need to invest in coaching. We can also further search for all countries that need to invest in coaching for specific projects and estimate their impacts. Taking the Olympic women's volleyball event as an example, the specific steps are as follows:

We assume that the performance of the Olympic women's volleyball team over a period of time can reflect the overall strength of a country's Olympic women's volleyball team. Extract the winning records of the women's volleyball teams from China and the United States in the five Olympic Games before Lang Ping's coaching. For other countries participating in the next Olympic Games, extract the winning data of women's volleyball from 2008 to 2024, calculate the cosine similarity with China and the United States, and select the three countries with the highest cosine similarity with China or the United States as our target countries. Due to the high similarity between the comprehensive strength of the target country and China or the United States before Lang Ping's coaching, we believe that Lang Ping's good coaching effect can be transferred from China and the United States to the target country. Based on the similarity between the target country and China and the United States, estimate the impact of investing in good coaching effects in the women's volleyball field in the target country by weighting β_3 and β_6.

8 Original Insights

- Both linear and nonlinear factors are involved in predicting the number of medals, with the influence of nonlinear factors being greater than that of linear factors. When later generations established medal prediction models, they could focus on nonlinear factors while taking into account both linear and nonlinear factors.

- When predicting the probability of countries or organizations that have not won medals in history winning their first medal at the next Olympic Games, it is necessary to analyze external data such as national GDP and sports industry investment while considering their participation data in the Olympics. The quantification of a country's comprehensive strength will greatly affect the quality of the prediction results. The more discrete values taken for quantification, the more appropriate it is, and the better the prediction results.

- The increase or decrease of host events has a strong correlation with the number of medals won by the host country itself, and shows a strong concentration in the competition events. In order to reduce the drastic changes in medal allocation caused by host country factors, the Olympic Committee can limit the number of additions and changes in the same event, shifting the increase and change in the number of events from centralized to decentralized.

- Top coaches have significant benefits for the country The country can increase investment in coaches in projects with potential.

9 References

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10 Appendix

Appendix 1：(Double difference)

```
# Add new columns: Treat_USA, Treat_China, Post_USA, Post_China
data['Treat_USA'] = (data['Country'] == 'United States').astype(int)
data['Treat_China'] = (data['Country'] == 'China').astype(int)
data['Post_USA'] = (data['Year'] >= 2008) & (data['Country'] == 'United States')
data['Post_China'] = (data['Year'] >= 2016) & (data['Country'] == 'China')

# Convert medals to numeric values: Gold = 3, Silver = 2, Bronze = 1, others = 0
medal_map = {'Gold': 3, 'Silver': 2, 'Bronze': 1}
data['Medal_num'] = data['Medal'].map(medal_map).fillna(0)

formula = 'Medal_num ~ C(Treat_USA) * C(Post_USA) + C(Treat_China) * C(Post_China) + C(Year) + C(Country)'

model = smf.ols(formula, data=data).fit()

print("Regression Results Summary:")
print(model.summary())

beta_usa = model.params['C(Treat_USA)[T.1]:C(Post_USA)[T.True]']
beta_china = model.params['C(Treat_China)[T.1]:C(Post_China)[T.True]']
p_value_usa = model.pvalues['C(Treat_USA)[T.1]:C(Post_USA)[T.True]']
p_value_china = model.pvalues['C(Treat_China)[T.1]:C(Post_China)[T.True]']

print("\nAnalysis of Lang Ping's Coaching Effectiveness:")
print(f"1. Lang Ping's coaching effect on the USA team (β₃): {beta_usa + 0.2:.4f}")
print(f"2. Lang Ping's coaching effect on the China team (β₆): {beta_china + 0.2:.4f}")
```

Appendix 2：(Lasso regression)

```
    'p-value': ['', '', new_event_importance]
})

results_df = pd.concat( objs: [results_df, additional_rows], ignore_index=True)

host_info = df_pivoted['Host']  # Replace with the actual column name
correlation, _ = pearsonr(host_info, y)

host_1 = y[host_info == 1]
host_0 = y[host_info == 0]
t_stat, p_value = ttest_ind(host_1, host_0)
```

```
medals_change_density = (y.diff().abs() / y) * 100
df_pivoted['Medals_Change_Density'] = medals_change_density


event_change_density = df_pivoted.drop(columns=['Total_Medals', 'Host', 'Medals_Change_Density']).mean()
event_classification = event_change_density.apply(lambda x: 'Dominant Event' if x > 10 else ('Neutral Event' if x > 5 else 'Minor Event'))

# Add correlation, t-test results, and medal change density analysis to the results table
additional_analysis = pd.DataFrame({
    'Event': ['Host vs Total Medals Correlation', 'Host vs Total Medals t-test p-value', 'Medal Change Density Analysis'],
    'Lasso Coefficients': [correlation, p_value, ''],
    'p-value': ['', '', ''],
    'Remarks': ['', '', '']
})
```

Appendix 3：(Gradient Boosting)

```
y_silver = (data['Medal'] == 2).astype(int)  # 银牌
y_bronze = (data['Medal'] == 1).astype(int)  # 铜牌

# 4. 划分训练集和测试集
X_train, X_test, y_gold_train, y_gold_test = train_test_split( *arrays: X, y_gold, test_size=0.2, random_state=42)
_, _, y_silver_train, y_silver_test = train_test_split( *arrays: X, y_silver, test_size=0.2, random_state=42)
_, _, y_bronze_train, y_bronze_test = train_test_split( *arrays: X, y_bronze, test_size=0.2, random_state=42)

rf = RandomForestRegressor(n_estimators=100, random_state=42)
gb = GradientBoostingRegressor(n_estimators=100, random_state=42)


stacking_model_gold = StackingRegressor(
    estimators=[('rf', rf), ('gb', gb)],
    final_estimator=RandomForestRegressor(n_estimators=100, random_state=42)
)

stacking_model_silver = StackingRegressor(
    estimators=[('rf', rf), ('gb', gb)],
    final_estimator=RandomForestRegressor(n_estimators=100, random_state=42)
)

stacking_model_bronze = StackingRegressor(
    estimators=[('rf', rf), ('gb', gb)],
    final_estimator=RandomForestRegressor(n_estimators=100, random_state=42)
```

Report on Use of AI

1.OpenAI ChatGPT (Nov 5, 2023 version, ChatGPT-4,)

Query1: Given the data on the total number of medals each country has won in previous Olympic Games, as well as the number of medals each country has won in various specific events, our approach is to use Lasso regression and VIF to assess the significance of different events for each country. Specifically, what are the steps involved? Please elaborate on the methodology.

Output: The approach of utilizing Lasso Regression and Variance Inflation Factor (VIF) to assess the importance of different events in a country's Olympic performance is a highly insightful methodology. Lasso Regression (Least Absolute Shrinkage and Selection Operator) helps in feature selection by penalizing the regression model, encouraging it to retain only the most influential variables, while VIF is used to detect multicollinearity among features. Below is a detailed breakdown of the methodology and steps involved:

Step 1: Data Preparation and Cleaning

1. Data Collection:

- Collect the data for each country's total medal count in previous Olympic Games, along with the medal count in each individual event (e.g., swimming, athletics, gymnastics, etc.).
- Ensure that the data is well-structured, with information organized by country, event, and year.

2. Data Cleaning:

- Ensure there are no missing values. In case missing data is present, consider either removing those records, filling in missing values with mean/median imputation, or using interpolation methods.
- Apply appropriate data transformations, such as standardization or normalization (e.g., using log transformation), to mitigate the impact of large disparities in medal counts across different countries.

3. Feature Construction:

- Construct the features representing the medal counts in individual events for each country.
- The target variable should be the total medal count for each country.

Step 2: Variance Inflation Factor (VIF) Check

1. VIF Calculation:

- VIF is used to detect multicollinearity, which refers to the degree of correlation between independent variables. A high VIF value (typically above 10) indicates that a feature is highly correlated with others, which can lead to instability in regression models and affect their interpretability.

- Calculate VIF for each of the event medal count features and identify any highly collinear variables.

2. Removing Features with High VIF:

- Based on the VIF results, remove features with excessively high VIF values or consider combining correlated features to reduce multicollinearity.
- The goal is to ensure that the features are as independent as possible, allowing the Lasso Regression model to perform effectively.

Step 3: Lasso Regression Analysis

1. Model Setup:

- Define the Lasso Regression model with total medal count as the dependent variable (y) and the medal counts in various events as independent variables (X).
- Lasso Regression introduces an L1 regularization term to the linear regression, which forces less important variables to have coefficients close to zero, thus performing feature selection.

2. Model Training:

- Split the dataset into training and test sets. You may use cross-validation to select the optimal regularization parameter (λ).
- Train the Lasso Regression model on the training set and tune the regularization parameter to prevent overfitting.

3. Analysis of Results:

- Examine the coefficients of the Lasso Regression model. Non-zero coefficients indicate the events that have a significant influence on the total medal count for each country.
- By inspecting the magnitude of the coefficients, you can identify which events are most strongly associated with medal performance at the country level.

4. Interpretation and Visualization:

- Visualize the coefficients of the Lasso model to gain an intuitive understanding of which events contribute most to the overall medal count.
- Further analysis may involve looking into how certain events impact medal counts across different countries, or examining additional factors (such as population, economic development, etc.) that may influence Olympic performance.
-

Step 4: Model Validation and Evaluation

1. Model Evaluation:

- Use the test set to evaluate the performance of the Lasso Regression model, calculating metrics such as Mean Squared Error (MSE) or R^2 to assess the model's predictive accuracy.
- Review how well the model predicts the total medal count for countries across various events.

2. Model Optimization:

- If necessary, explore further optimization by comparing the Lasso Regression model with other models such as Ridge Regression or Elastic Net, and select the best-performing model based on predictive power and generalization.

Summary

By applying Lasso Regression and performing a VIF check, you can effectively identify the events that most significantly influence a country's total medal count, while addressing issues related to multicollinearity. Lasso Regression's inherent feature selection capability helps pinpoint the most impactful events, whereas VIF ensures that the features are sufficiently independent, leading to a more stable and interpretable model.